

Skill 1

```
nishanth@LAPTOP-6Q3ON07H:~$ pwd
/home/nishanth
nishanth@LAPTOP-6Q3ON07H:~$ ls
OSLAB  OSLab  demo  demo.c  fork  forklab  frok  lab  practical1  practical2  process  process.c  waitlab
nishanth@LAPTOP-6Q3ON07H:~$ mkdir LinuxLab
nishanth@LAPTOP-6Q3ON07H:~$ cd Linux Lab
-bash: cd: too many arguments
nishanth@LAPTOP-6Q3ON07H:~$ cd LinuxLab
nishanth@LAPTOP-6Q3ON07H:~/LinuxLab$ echo "Hello Linux" > file1.txt
nishanth@LAPTOP-6Q3ON07H:~/LinuxLab$ cat file1.txt
Hello Linux
nishanth@LAPTOP-6Q3ON07H:~/LinuxLab$ mv file2.txt renamed.txt
mv: cannot stat 'file2.txt': No such file or directory
nishanth@LAPTOP-6Q3ON07H:~/LinuxLab$ echo "Hello Linux" > file1.txt
nishanth@LAPTOP-6Q3ON07H:~/LinuxLab$ cp file1.txt file2.txt
nishanth@LAPTOP-6Q3ON07H:~/LinuxLab$ mv file2.txt renamed.txt
nishanth@LAPTOP-6Q3ON07H:~/LinuxLab$ find . -name "renamed.txt" ./renamed.txt
find: paths must precede expression: `./renamed.txt'
find: possible unquoted pattern after predicate `-name'?
nishanth@LAPTOP-6Q3ON07H:~/LinuxLab$ find . -name "renamed.txt"
./renamed.txt
```

```
nishanth@LAPTOP-6Q3ON07H:~/LinuxLab$ history
```

```
1  sudo apt update
2  sudo apt install build-essential
3  gcc --version
4  mkdir OSLab
5  cd OSLab
6  gedit hello.c
7  sudo apt install gedit
8  gedit hello.c
9  sudo apt update
10 sudo apt install gcc
11 gcc --version
12 ls
13 gedit hello.c
14 mkdir OSLab
15 cd OSLab
16 gedit hello.c
17 nano process.c
18 ls
19 gcc process.c -o process
20 ./process
21 uname
22 lscpu
23 lsblk
24 ps
25 top
26 echo $shell
27 echo $SHELL
28 nano demo.c
29 gcc demo.c -o demo
30 ./demo
31 help
32 uname -a
33 mkdir fork
34 cd fork
35 nano program1.c
36 gcc program1.c -o program1
37 nano program1.c
38 gcc program1.c -o program1
39 ./program1
40 nano program2
41 gcc program1.c -o program2
42 ./program2
43 ./program1
44 nano program1.c
45 ls
46 rm program2
47 ls
48 nano program2.c
49 gcc program2.c -o program2
```

```
49 gcc program2.c -o program2
50 ./program2
51 ls
52 nano program3.c
53 gcc program3.c -o program3
54 ./program3
55 nano program4.c
56 gcc program4.c -o program4
57 ./program4
58 nano program5.c
59 gcc program5.c -o program5
60 ./program5
61 nano program6.c
62 gcc program6 -o program6
63 gcc program6.c -o program6
64 ./program6
65 ./program4
66 nano program7.c
67 gcc program7.c -o program7
68 ./program7
69 nano program8.c
70 gcc program8.c -o program8
71 ./program8
72 cd ~/fork
73 ls
74 cat program1.c
75 ./program1
76 cat program2.c
77 ./program2
78 cat progra3.c
79 cat program3.c
80 ./program3
81 cat program4.c
82 ./program4
83 cat program5.c
84 ./program5
85 cat program6.c
86 ./program6
87 cat program7.c
88 ./program7
89 cat program8.c
90 ./program8
91 history
92 history | sed -n '33,71p'
93 nano program9.c
94 gcc program9.c -o program9
95 ./program9
96 history
97 cd ~/fork
98 ls
```

```
100 ./program1
101 pwd
102 cd OSLAB
103 mkdir OSLAB
104 cd OSLAB
105 mkdir exec
106 cd exec
107 nano program1.c
108 gcc program1.c -o progarm1
109 ./program1
110 gcc program1.c -o program1
111 ./progarm1.c
112 ./program1
113 nano program2.c
114 gcc program2.c -o program2
115 ./program2
116 nano program3.c
117 gcc program3.c -o program3
118 ./program3
119 nano program4.c
120 gcc program4.c -o program4
121 ./program4
122 nano program5.c
123 gcc program5.c -o program5
124 ./program5
125 cat program1.c
126 ./program1
127 cat program2.c
128 ./program2
129 cat program3.c
130 ./program3
131 cat program4.c
132 ./program4
133 cat program5.c
134 ./program5
135 mkdir practical1
136 cd practical1
137 nano program1.c
138 gcc program1.c -o program1
139 ./program1.c
140 ./program1
141 cat program1.c
142 nano program2.c
143 gcc program2.c -o program2
144 ./program2
145 nano program3.c
146 gcc program3.c -o program3
147 ./program3
148 nano program4.c
149 gcc program4.c -o program4
```

```
159 ls
160 history
161 mkdir practical2
162 cd practical2
163 pwd
164 mkdir practical2
165 cd practical2
166 nano program1.c
167 gcc program1.c -o program1
168 ./program2
169 ./program1
170 nano program2.c
171 gcc program2.c -o program2
172 ./program2
173 echo "Hello, this is test content." > source.txt
174 ls
175 ./program2
176 nano program3.c
177 gcc program3.c -o program3
178 ./program3
179 nano program4.c
180 gcc program4.c -o program4
181 ./program4
182 history
183 mkdir fork process
184 mkdir frok
185 cd fork
186 nano program1.c
187 pwd
188 mkdir forklab
189 cd forklab
190 nano program1.c
191 gcc program1.c -o program1
192 ./program1
193 nano program2.c
194 gcc program2.c -o program2
195 ./program2
196 mkdir fork lab
197 mkdir forklab
198 cd forklab
199 ls
200 nano program3.c
201 gcc program3.c -o program3
202 ./program3
203 mkdir waitlab
204 cd waitlab
205 nano program1.c
206 gcc program1.c -o program
207 ./program1
208 gcc program1.c -o program1
```

```
209 ./program1
210 nano program2.c
211 gcc program2.c -o program2
212 ./program2
213 nano zombie.c
214 gcc zombie.c -o zombie
215 ./zombie
216 nano no_zombie.c
217 gcc no_zombie.c -o no_zombie
218 ./no_zombie
219 ls
220 pwd
221 cd
222 mkdir
223 mkdir test folder
224 cd test folder
225 rmdir test folder
226 cp
227 cp file1.txt file2.txt
228 mv
229 cd fork
230 mv
231 mv copy.txt newfile.txt
232 pwd
233 ls
234 mkdir LinuxLab
235 cd Linux Lab
236 cd LinuxLab
237 echo "Hello Linux" > file1.txt
238 cat file1.txt
239 mv file2.txt renamed.txt
240 echo "Hello Linux" > file1.txt
241 cp file1.txt file2.txt
242 mv file2.txt renamed.txt
243 find . -name "renamed.txt" ./renamed.txt
244 find . -name "renamed.txt"
245 history
```

```

nishanth@LAPTOP-6Q3ON07H:~/LinuxLab$ cd ..
nishanth@LAPTOP-6Q3ON07H:~$ rm -r LinuxLab
nishanth@LAPTOP-6Q3ON07H:~$ ls
OSLAB  demo    fork    frok    practical1  process    waitlab
OSLab  demo.c  forklab  lab    practical2  process.c
nishanth@LAPTOP-6Q3ON07H:~$ sudo apt update
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [371 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [89.4 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [32.7 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [150 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [48.3 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [321 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [61.2 kB]
Hit:12 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [432 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [105 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [62.0 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [223 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [72.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [161 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [331 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [63.0 kB]
Fetched 2843 kB in 6s (510 kB/s)
94 packages can be upgraded. Run 'apt list --upgradable' to see them.
nishanth@LAPTOP-6Q3ON07H:~$ sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 94
nishanth@LAPTOP-6Q3ON07H:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
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This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

nishanth@LAPTOP-6Q3ON07H:~$ sudo apt install git
git is already the newest version (1:2.53.0-1ubuntu1).
git set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 94
nishanth@LAPTOP-6Q3ON07H:~$ git --version
git version 2.53.0
nishanth@LAPTOP-6Q3ON07H:~$ mkdir newProject
nishanth@LAPTOP-6Q3ON07H:~$ cd newProject

```

```

nishanth@LAPTOP-6Q3ON07H:~$ git --version
git version 2.53.0
nishanth@LAPTOP-6Q3ON07H:~$ mkdir newProject
nishanth@LAPTOP-6Q3ON07H:~$ cd newProject
nishanth@LAPTOP-6Q3ON07H:~/newProject$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/nishanth/newProject/.git/
nishanth@LAPTOP-6Q3ON07H:~/newProject$ mkdir src
nishanth@LAPTOP-6Q3ON07H:~/newProject$ mkdir include
nishanth@LAPTOP-6Q3ON07H:~/newProject$ mkdir bin
nishanth@LAPTOP-6Q3ON07H:~/newProject$ mkdir docs
nishanth@LAPTOP-6Q3ON07H:~/newProject$ ls
bin docs include src
nishanth@LAPTOP-6Q3ON07H:~/newProject$ echo "Hello World" > src/main.c
nishanth@LAPTOP-6Q3ON07H:~/newProject$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    src/

nothing added to commit but untracked files present (use "git add" to track)
nishanth@LAPTOP-6Q3ON07H:~/newProject$ git add .
nishanth@LAPTOP-6Q3ON07H:~/newProject$ git config --global user.name "Your Name"
nishanth@LAPTOP-6Q3ON07H:~/newProject$ git config --global user.email "you@gmail.com"
nishanth@LAPTOP-6Q3ON07H:~/newProject$ git commit -m "First Commit"
git: 'comit' is not a git command. See 'git --help'.

The most similar command is
    commit
nishanth@LAPTOP-6Q3ON07H:~/newProject$ ls
bin docs include src

```

```
nishanth@LAPTOP-6Q3ON07H:~/newProject$ sudo apt install strace
Installing:
    strace

Installing dependencies:
    libunwind8

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 94
  Download size: 737 kB
  Space needed: 2660 kB / 1024 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunwind8 amd64 1.8.3-0ubuntu1 [59.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 strace amd64 6.19+ds-0ubuntu5 [678 kB]
Fetched 737 kB in 2s (431 kB/s)
Selecting previously unselected package libunwind8:amd64.
(Reading database ... 48265 files and directories currently installed.)
Preparing to unpack .../libunwind8_1.8.3-0ubuntu1_amd64.deb ...
Unpacking libunwind8:amd64 (1.8.3-0ubuntu1) ...
Selecting previously unselected package strace.
Preparing to unpack .../strace_6.19+ds-0ubuntu5_amd64.deb ...
Unpacking strace (6.19+ds-0ubuntu5) ...
Setting up libunwind8:amd64 (1.8.3-0ubuntu1) ...
Setting up strace (6.19+ds-0ubuntu5) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
nishanth@LAPTOP-6Q3ON07H:~/newProject$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7ffc60755fd0 /* 26 vars */) = 0
brk(NULL)                               = 0x5e94f0648000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x77b468334000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=27889, ...}) = 0
mmap(NULL, 27889, PROT_READ, MAP_PRIVATE, 3, 0) = 0x77b46832d000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x77b4682fa000
mmap(0x77b468301000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x77b468301000
mmap(0x77b468322000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x77b468322000mmap(0x77b468329000, 8192, PROT_
8329000
mmap(0x77b46832b000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x77b46832b000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x77b4682cc000
mmap(0x77b4682d0000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x77b4682d0000
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close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\250\2\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x77b4681a6000
mmap(0x77b4681b8000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x77b4681b8000
mmap(0x77b468258000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x77b468258000
mmap(0x77b4682ca000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x77b4682ca000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\250\2\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x77b467e00000
mmap(0x77b467e28000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x77b467e28000
mmap(0x77b467fc0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c0000) = 0x77b467fc0000
mmap(0x77b46800e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20d000) = 0x77b46800e000
mmap(0x77b468014000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x77b468014000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=699056, ...}) = 0
mmap(NULL, 701112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x77b4680fa000
mmap(0x77b4680fd000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3000) = 0x77b4680fd000
mmap(0x77b468179000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7f000) = 0x77b468179000
mmap(0x77b4681a4000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xa9000) = 0x77b4681a4000
close(3) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x77b4680f8000
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x77b4680f5000
arch_prctl(ARCH_SET_FS, 0x77b4680f5b00) = 0
set_tid_address(0x77b4680f6128) = 1226
set_robust_list(0x77b4680f5de0, 24) = 0
rseq(0x77b4680f56a0, 0x20, 0, 0x53053053) = 0
mprotect(0x77b46800e000, 16384, PROT_READ) = 0
mprotect(0x77b4681a4000, 4096, PROT_READ) = 0
mprotect(0x77b4682ca000, 4096, PROT_READ) = 0
mprotect(0x77b4682f8000, 4096, PROT_READ) = 0
mprotect(0x77b468329000, 4096, PROT_READ) = 0
mprotect(0x5e94ce948000, 1462272, PROT_READ) = 0
mprotect(0x77b468379000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
getrandom("\x22\x4f\x32\x16\x61\x20\x31\x06", 8, GRND_NONBLOCK) = 8
munmap(0x77b46832d000, 27889) = 0
statfs("/sys/fs/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0,
ID|ST_NODEV|ST_NOEXEC|ST_NOATIME}) = 0
statfs("/selinux", 0x7ffc361ae0d0) = -1 ENOENT (No such file or directory)
brk(NULL) = 0x5e94f0648000

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statfs("/selinux", 0x7ffc361ae0d0) = -1 ENOENT (No such file or directory)
brk(NULL) = 0x5e94f0648000
brk(0x5e94f0669000) = 0x5e94f0669000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\ttsysfs\nnodev\ttmpfs\nnodev\tbdev\n...", 1024) = 442
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.18.33.2-... 1024) = 1024
read(3, "e 0 0\ndevpts /dev/pts devpts rw,... 1024) = 1024
read(3, ",access=client,msize=65536,trans"... 1024) = 1024
read(3, "des=1024,mode=700,noswap 0 0\ntmp"... 1024) = 262
read(3, "", 1024) = 0
close(3) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Timeout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x77b467e45cb0}, 8) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "5e94cdfdf000-5e94ce0d1000 r--p 0"... 1024) = 1024
read(3, "30 14176 /u"... 1024) = 1024
read(3, ":30 14388 /"... 1024) = 1024
read(3, " /usr/lib/x86_64-linux-gnu"... 1024) = 1024
read(3, "31 /usr/lib"... 1024) = 561
close(3) = 0
sched_getaffinity(1226, 32, [0 1 2 3 4 5 6 7 8 9 10 11]) = 8
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x77b468331000
mprotect(0x77b468331000, 4096, PROT_NONE) = 0
sigaltstack({ss_sp=0x77b468332000, ss_flags=0, ss_size=8192}, NULL) = 0
gettid() = 1226
rt_sigaction(SIGSEGV, {sa_handler=0x5e94ce3040c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x77b468331000}, 8) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=0x5e94ce3040c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x77b468331000}, 8) = 0
prctl(PR_GET_AUXV, 0x7ffc361adbc0, 0x200, 0, 0) = 448
statx(AT_FDCWD, "/usr/share/coreutils/locales/lc", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_size=0, stx_block_size=0, stx_object_id=0, stx_pad=0}, 0) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/uucore", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_size=0, stx_block_size=0, stx_object_id=0, stx_pad=0}, 0) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_size=0, stx_block_size=0, stx_object_id=0, stx_pad=0}, 0) = 0
read(3, "# Common strings shared across a"... 4080) = 4080
read(3, "", 32) = 0

```



```

rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7e3ec5445cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
openat(AT_FDCWD, "/proc/self/maps", 0_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "630015ddd000-630015ecf000 r--p 0"..., 1024) = 1024
read(3, "20d000 08:30 14176"..., 1024) = 1024
read(3, "0012000 08:30 14388"..., 1024) = 1024
read(3, " /usr/lib/x86_64"..., 1024) = 1024
read(3, " 08:30 13931"..., 1024) = 571
close(3) = 0
sched_getaffinity(1250, 32, [0 1 2 3 4 5 6 7 8 9 10 11]) = 8
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7e3ec58e2000
mprotect(0x7e3ec58e2000, 4096, PROT_NONE) = 0
sigaltstack({ss_sp=0x7e3ec58e3000, ss_flags=0, ss_size=8192}, NULL) = 0
gettid() = 1250
rt_sigaction(SIGSEGV, {sa_handler=0x6300161020c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7e3ec5445cb0}, NULL, 8) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=0x6300161020c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7e3ec5445cb0}, NULL, 8) = 0
prctl(PR_GET_AUXV, 0x7fff5e85a710, 0x200, 0, 0) = 448
statx(AT_FDCWD, "/usr/share/coreutils/locales/date", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=0, stx_size=0, stx_blksize=0, stx_blocks=0, stx_rdev=0, stx_ino=0, stx_nlink=0, stx_uid=0, stx_gid=0, stx_mnt_id=0, stx_mnt_flags=0, stx_magic=0, stx_flags=0}, 0) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/uucore", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=0, stx_size=0, stx_blksize=0, stx_blocks=0, stx_rdev=0, stx_ino=0, stx_nlink=0, stx_uid=0, stx_gid=0, stx_mnt_id=0, stx_mnt_flags=0, stx_magic=0, stx_flags=0}, 0) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", 0_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, stx_blksize=0, stx_blocks=0, stx_rdev=0, stx_ino=0, stx_nlink=0, stx_uid=0, stx_gid=0, stx_mnt_id=0, stx_mnt_flags=0, stx_magic=0, stx_flags=0}, 0) = 0
read(3, "# Common strings shared across a"..., 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/date", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=0, stx_size=0, stx_blksize=0, stx_blocks=0, stx_rdev=0, stx_ino=0, stx_nlink=0, stx_uid=0, stx_gid=0, stx_mnt_id=0, stx_mnt_flags=0, stx_magic=0, stx_flags=0}, 0) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/date/en-US.ftl", 0_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
brk(0x6300459d4000) = 0x6300459d4000
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7e3ec5445cb0}, {sa_handler=SIG_IGN, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7e3ec5445cb0}, {sa_handler=0x6300161020c0, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7e3ec5445cb0}, {sa_handler=0x6300161020c0, sa_mask=[], sa_flags=0}, 0) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|...}) = 0
openat(AT_FDCWD, "/usr/share/zoneinfo", 0_RDONLY|O_NONBLOCK|O_CLOEXEC|O_DIRECTORY) = 3
fstat(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
getdents64(3, 0x6300459b8930 /* 25 entries */, 32768) = 760
getdents64(3, 0x6300459b8930 /* 0 entries */, 32768) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/share/zoneinfo/Pacific", 0_RDONLY|O_NONBLOCK|O_CLOEXEC|O_DIRECTORY) = 3
fstat(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
getdents64(3, 0x6300459b8930 /* 43 entries */, 32768) = 1312
getdents64(3, 0x6300459b8930 /* 0 entries */, 32768) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/share/zoneinfo/Etc", 0_RDONLY|O_NONBLOCK|O_CLOEXEC|O_DIRECTORY) = 3

```



```

nishanth@LAPTOP-6Q30N07H:~$ cd newProject
nishanth@LAPTOP-6Q30N07H:~/newProject$ ls
bin      child    docs    exec.c  fork.c  include  process.c  shell.c
bye.c    child.c  exec    fork    hello   process  shell      src
nishanth@LAPTOP-6Q30N07H:~/newProject$ cat exec.c
#include<stdio.h>
#include<unistd.h>

int main()
{
    printf("Before exec\n");

    execl("/bin/ls", "ls", NULL);

    printf("After exec");

    return 0;
}
nishanth@LAPTOP-6Q30N07H:~/newProject$ ./exec
Before exec
bin      child    docs    exec.c  fork.c  include  process.c  shell.c
bye.c    child.c  exec    fork    hello   process  shell      src

```

```

nishanth@LAPTOP-6Q30N07H:~/newProject$ ./exec
Before exec
bin      child    docs    exec.c  fork.c  include  process.c  shell.c  wait
bye.c    child.c  exec    fork    hello   process  shell      src      wait.c
nishanth@LAPTOP-6Q30N07H:~/newProject$ cat bye.c
#include <stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
nishanth@LAPTOP-6Q30N07H:~/newProject$ ./bye
-bash: ./bye: No such file or directory
nishanth@LAPTOP-6Q30N07H:~/newProject$ ./hello.c
Hello Linux

```

```
nishanth@LAPTOP-6Q3ON07H:~/newProject$ ./bye
-bash: ./bye: No such file or directory
nishanth@LAPTOP-6Q3ON07H:~/newProject$ ./hello
Hello Linux
nishanth@LAPTOP-6Q3ON07H:~/newProject$ cat fork.c
#include<stdio.h>
#include<unistd.h>

int main()
{
    fork();

    printf("Hello\n");

    return 0;
}
```

```
nishanth@LAPTOP-6Q3N07H:~/newProject$ ./fork
Hello
Hello
nishanth@LAPTOP-6Q3N07H:~/newProject$ |
```

```
nishanth@LAPTOP-6Q30N07H:~/newProject$ cat child.c
#include<stdio.h>
#include<unistd.h>

int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)
    {
        printf("Child PID=%d\n",getpid());
        printf("Parent PID=%d\n",getppid());
    }
    else
    {
        printf("Parent PID=%d\n",getpid());
    }

    return 0;
}
nishanth@LAPTOP-6Q30N07H:~/newProject$ ./child
Parent PID=9523
Child PID=9524
Parent PID=8827
```

```
nishanth@LAPTOP-6Q30N07H:~/newProject$ cat make
cat: make.c: No such file or directory
nishanth@LAPTOP-6Q30N07H:~/newProject$ nano wai
nishanth@LAPTOP-6Q30N07H:~/newProject$ cat wait
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>

int main()
{
    if(fork()==0)
    {
        printf("Child\n");
    }
    else
    {
        wait(NULL);

        printf("Parent\n");
    }

    return 0;
}
nishanth@LAPTOP-6Q30N07H:~/newProject$ ./wait
```

```
Child
Parent
nishanth@LAPTOP-6Q30N07H:~/newProject$ |
```

```
nishanth@LAPTOP-6Q30N07H:~/newProject$ cat process.c
#include<stdio.h>
#include<unistd.h>

int main()
{
    fork();
    fork();

    printf("PID=%d\n",getpid());

    return 0;
}
nishanth@LAPTOP-6Q30N07H:~/newProject$ ./process
```

PID=12345

PID=12347

PID=12346

PID=12348

nishanth@LAPTOP-6Q30N07H:~/newProject\$ |